INCH-POUND
MIL-PRF-3098/4G
w/AMENDMENT 1
15 July 2019
SUPERSEDING
MIL-PRF-3098/4G
5 March 2010

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR19/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 0.8 MHz to 20 MHz; fundamental; noncontrolled; series resonance.

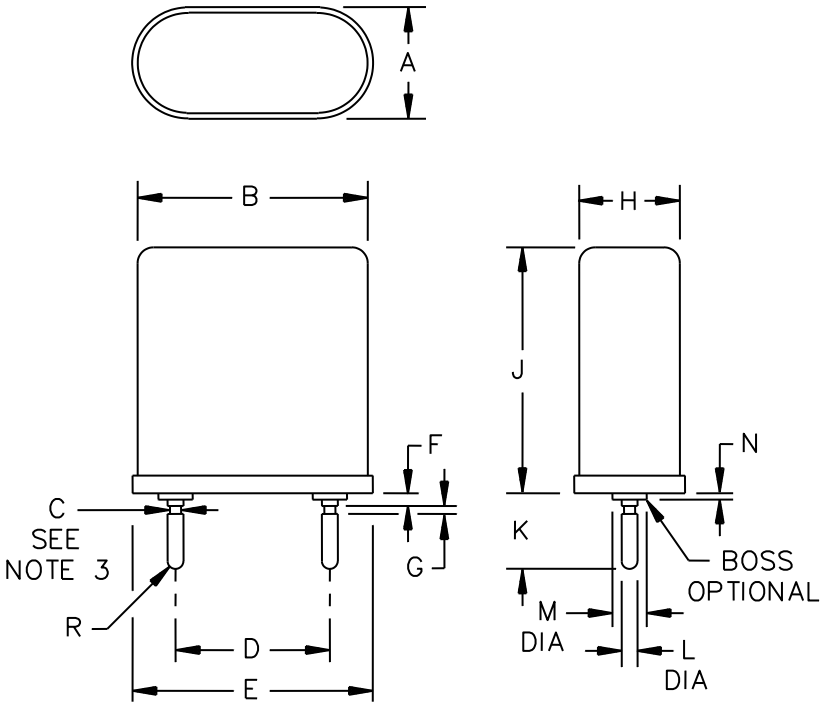


FIGURE 1. Crystal unit - CR19/U.

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.352	---	8.94
B	---	.725	---	18.42
C	.030	.037	0.76	0.94
D	.478	.494	12.14	12.55
E	---	.757	---	19.23
F	.030	.040	0.76	1.02
G	.015	.025	0.38	0.64
H	---	.317	---	8.05
J	---	.775	---	19.69
K	.223	.248	5.66	6.30
L	.048	.052	1.22	1.32
M	.075	.141	1.91	3.58
N	.015	.025	0.38	0.64

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR19/U - Continued.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 0.8 MHz to 20 MHz, inclusive.

Frequency tolerance: ±50 parts per million (ppm).

Equivalent resistance: See table I.

Mode of oscillation: Fundamental.

Resonance: Series.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Capacitance, shunt: 7 pF, maximum.

Shock (specified pulse):

Frequency change permitted: 0.8 MHz to 2.0 MHz: ±10 ppm
2.0+ MHz to 20 MHz: ± 5 ppm

Equivalent resistance change permitted: 0.8 MHz to 2.0 MHz: ±15 percent
2.0+ MHz to 20 MHz: ±10 percent

Thermal shock:

Frequency change permitted: 0.8 MHz to 2.0 MHz: ±10 ppm
2.0+ MHz to 20 MHz: ± 5 ppm

Equivalent resistance change permitted: 0.8 MHz to 2.0 MHz: ±15 percent
2.0+ MHz to 20 MHz: ±10 percent

Vibration: MIL-STD-202-201.

Frequency change permitted: 0.8 MHz to 2.0 MHz: ±10 ppm
2.0+ MHz to 20 MHz: ± 5 ppm

Equivalent resistance change permitted: 0.8 MHz to 2.0 MHz: ±15 percent
2.0+ MHz to 20 MHz: ±10 percent

Low temperature storage:

Resistance: See table I.

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Temperature run:

Frequency change permitted: 0.8 MHz to 2.0 MHz: ±10 ppm
2.0+ MHz to 20 MHz: ± 5 ppm

Equivalent resistance change permitted: 0.8 MHz to 2.0 MHz: ±15 percent
2.0+ MHz to 20 MHz: ±10 percent

Aging:

Frequency change permitted: ±5 ppm.

TABLE I. Equivalent resistance.

Frequency range, inclusive	Maximum resistance	Frequency range, inclusive	Maximum resistance
MHz	Ohms	MHz	Ohms
0.80 to 0.85	520	2.12+ to 2.25	150
0.85+ to 0.90	480	2.25+ to 2.60	130
0.90+ to 1.00	440	2.60+ to 3.00	90
1.00+ to 1.12	400	3.00+ to 3.40	70
1.12+ to 1.25	380	3.40+ to 3.75	52
1.25+ to 1.37	340	3.75+ to 4.00	45
1.37+ to 1.50	300	4.00+ to 5.00	37
1.50+ to 1.62	280	5.00+ to 7.00	25
1.62+ to 1.75	250	7.00+ to 10.00	20
1.75+ to 1.87	220	10.00+ to 15.00	18
1.87+ to 2.00	190	15.00+ to 20.00	15
2.00+ to 2.12	170		

Part or Identifying Number (PIN): CR19/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Referenced documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202-201

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Custodians:
Army - CR
Navy - EC
DLA - CC

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-2019-006)

Review activities:
Army - AR, MI
Navy - AS, MC, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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